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EE431 - Digital Image Processing

2024-2025 Spring

Instructor: Asst. Prof. Dr. Cem Kalyoncu

Final

30 May 2025

ID: _____

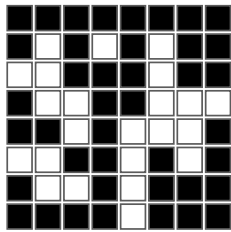
Name: _____

Electronic devices other than standard scientific calculators are not allowed during the exam. Mobile phones are **not** allowed within the exam room. Try to answer questions in the given space. Exam duration is **70** minutes and it is out of **41** points. Good luck.

$$\text{Entropy} = - \sum \log P_i P_i$$

Learning outcomes (LO) and program outcomes (ME) are given next to the questions.

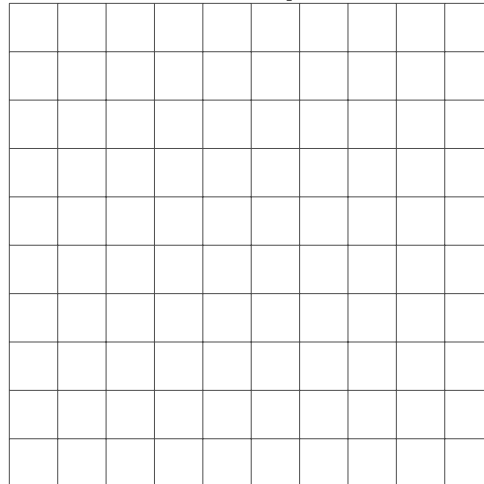
Question 1: Perform closing to the following image using the first kernel followed by a hit and miss transform using the second kernel to the closed result. First step image size is larger to help you work with out of image pixels. Cross any pixel that is true/white. (LO4, ME1, ME2) **[12 points]**



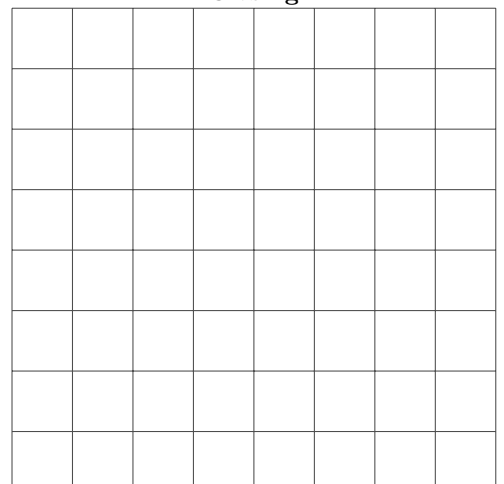
Kernel

0	0	0
1	1	1
0	0	0

First step

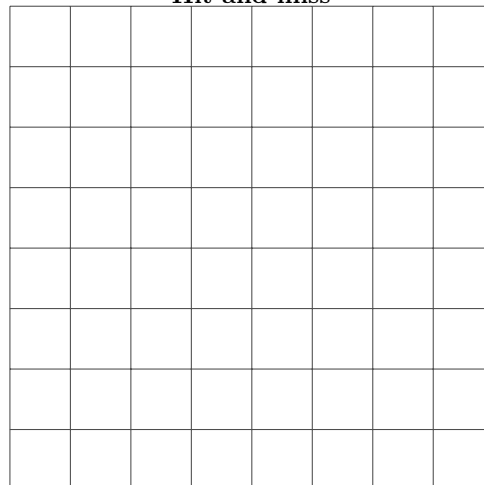


Closing



0	1	0
-1	1	0
0	0	0

Hit and miss



Question 2: The following 4 bit image is given. (LO4, LO5, ME2, ME4)

a. Calculate the entropy of this image as bits/sample. [4 points]

7	2	1	1	2	7
10	5	2	2	5	10
15	10	5	5	2	2
12	15	10	5	5	5
12	12	15	5	7	5
10	15	2	2	7	10

b. What is the theoretical maximum compression ratio? **[2 points]**

c. Draw the Huffman tree for the image to the next page. [7 points]

d. Write the Huffman encoded image in binary. **[3 points]**

Output

[illegible]

e. Calculate compression ratio. [2 points]

Question 3: Write a program that will segment an image I using saturation channel. A simple saturation formula of $S = \frac{\max(R,G,B) - \min(R,G,B)}{\text{avg}(R,G,B)}$ is given, if average is 0 then use 0 for S . The input image I should be segmented to output image O using average saturation of the input image as the threshold. For one point bonus point, calculate and clearly state what is the maximum saturation that can be obtained by this formula. *Just in case, write in two columns.* (LO3, LO4, ME3) **[11 point]**